



SEMESTER 1 EXAMINATIONS 2023/2024

MODULE: CA318 - Advanced Algorithms and A.I. Search

PROGRAMME(S):

COMSCI	BSc in Computer Science
ECSAO	Study Abroad (Engineering & Computing)
CASE	BSc in Computer Applications (Sft.Eng.)

YEAR OF STUDY: 3,0

EXAMINER(S):

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TIME ALLOWED: 2 Hours

INSTRUCTIONS: See inside.

PLEASE DO NOT TURN OVER THIS PAGE UNTIL YOU ARE INSTRUCTED TO DO SO.
The use of programmable or text storing calculators is expressly forbidden.

There are no additional requirements for this paper.

Multiple choice

Answer all 20 multiple-choice questions. (2 marks each.) Total 40 marks.

No negative marking.

Only the filled-in boxes will be looked at. No additional commentary or analysis will be read.

Remember to hand up the multiple-choice answer sheet!

1. In neural networks, let $\frac{\delta E}{\delta w}$ be the change in error as we change a weight. Let c be a positive constant. Then a rule to reduce error would be:

1. $W := c \frac{\delta E}{\delta w}$

2. $W := W - c \frac{\delta E}{\delta w}$

3. $W := W + c \frac{\delta E}{\delta w}$

4. $W := -c \frac{\delta E}{\delta w}$

2. One of these is *not* a property of neural networks:
1. It can be used to represent any normal non-linear, continuous, non-chaotic function.
 2. Input is compared to previous Input, and "nearby" Input should produce "nearby" Output.
 3. It can be used to represent a function from integer-valued multi-dimensional input to integer-valued multi-dimensional output.
 4. Given a network architecture, an Input leads to an Output after a *fixed* number of calculations.
 5. It can be used to represent a function from real-valued single-dimensional input to real-valued single-dimensional output.
 6. It can be used to represent a function from real-valued multi-dimensional input to real-valued multi-dimensional output.
 7. As learning progresses, the *size* of the network remains *fixed*.
 8. It can be used to represent any normal linear, continuous, non-chaotic function.

3. Define a perceptron that receives x and y as input and fires for these points:

$$ax - by \leq -c$$

1. weights: a, b , threshold: $-c$
2. weights: $-a, b$, threshold: $-c$
3. weights: a, b , threshold: c
4. weights: $-a, b$, threshold: c
5. weights: $a, -b$, threshold: c
6. weights: $-a, -b$, threshold: $-c$
7. weights: $a, -b$, threshold: $-c$
8. weights: $-a, -b$, threshold: c

4. We do machine evolution as follows:

Start with 100 random solutions.

Repeat:

Test all 100 for fitness.

Pick the 10 best.

Make 10 copies of each to construct the next generation.

Loop again.

The problem with this is:

1. There is no problem.
2. We should pick according to probabilities rather than an absolute number.
3. 100 is too small.
4. It will end up focused on one solution.
5. Too many solutions reproduce.
6. 10 is too small.
7. We lose the best solutions after finding them.
8. No new solution is ever made.

5. One of these is *known* to be false:

1. In the future, immortal humans will travel to distant stars.
2. In the future, intelligent machines will exterminate the human race.
3. AI is possible.
4. AI (the construction of intelligent machines) is impossible.
5. Machines have beaten world-class humans at tennis.
6. Machines have beaten world-class humans at chess.

7. In the future, humans will live forever on earth in artificial bodies, regularly backing up their brains in order to cheat death.

6. An upper bound on the *game-tree complexity* of Xs and Os is:

1. $9!$
2. $3!$
3. 3^9
4. 3^3
5. 9^9
6. 9^3

7. Consider a heuristic for human mate choice:

Have a relationship with n partners of fitness $> f$
Marry the next partner you find of fitness $> g$

where fitness is ranked by the individual out of 10.

Which of the following values for n, f, g (respectively) makes the most sense?

1. 0, 0, 8
2. 10, 8, 3
3. 0, 5, 8
4. 10, 0, 8
5. 10, 5, 8
6. 0, 8, 3

8. A multi-layer neural network has 1 input node, 12 hidden nodes, and 1 output node. The total number of parameters to be learnt (weights and thresholds) in the network is:

1. 38.
2. 1.
3. 13.
4. 24.
5. 12.
6. 14.
7. 37.

9. The set of inequalities that define a perceptron to implement OR are:

1. $0 \leq t, w_1 > t, w_2 > t$
2. $0 \geq t, w_1 < t, w_2 < t$
3. $0 > t, w_1 \leq t, w_2 \leq t$
4. $0 < t, w_1 \geq t, w_2 \geq t$
5. $0 \leq t, w_1 \geq t, w_2 \geq t$
6. $0 \geq t, w_1 \leq t, w_2 \leq t$
7. $0 > t, w_1 < t, w_2 < t$
8. $0 < t, w_1 > t, w_2 > t$

10. Neural Networks and GAs can both be seen as hill-climbing on an unknown landscape by taking a finite number of samples. We consider E , the error in

our solution, and $\frac{\delta E}{\delta w}$ the direction in which to change our solution. Which of the following is the case:

1. For Neural Networks, E is known and $\frac{\delta E}{\delta w}$ is known. For GAs, E is known but $\frac{\delta E}{\delta w}$ is unknown.
2. For Neural Networks, E is known and $\frac{\delta E}{\delta w}$ is known. For GAs, E is known and $\frac{\delta E}{\delta w}$ is known.
3. For Neural Networks, E is known but $\frac{\delta E}{\delta w}$ is unknown. For GAs, neither E nor $\frac{\delta E}{\delta w}$ are known.
4. For Neural Networks, E is known and $\frac{\delta E}{\delta w}$ is known. For GAs, neither E nor $\frac{\delta E}{\delta w}$ are known.

11. In AI search, the following heuristic:

```
Initialise with 10 random solutions.  
Repeat forever  
{  
  Pick the best solution.  
  Make 10 new solutions by making small or no changes to it.  
}
```

might be a bad heuristic because:

1. It tends to converge on local optima.
2. 10 is too small.
3. It is not a heuristic.
4. It tends to converge on global optima.
5. 10 is too large.
6. Small changes may break the best solution.
7. It tends to lose good solutions once found.
8. It is not a bad heuristic.

12. In machine evolution, let the probability of an individual reproducing be *its fitness divided by the sum of fitnesses of all individuals*. Then:

1. Fitness must be ≥ 0
2. Fitness must be > 0
3. Fitness must be ≤ 0
4. Fitness can have any value except zero.
5. Fitness must be < 0
6. Fitness can have any value.

13. Playing chess requires what heuristics?

1. Heuristic search only.
2. Heuristic search, and heuristic evaluation function.
3. No heuristics.
4. Heuristic evaluation function only.

14. Consider a heuristic for human mate choice:

```
Have a relationship with n partners of fitness > f  
Marry the next partner you find of fitness > g
```

where fitness is ranked by the individual out of 10.

Which of the following values for n, f, g (respectively) makes the most sense?

1. 100, 0, 8
2. 100, 8, 3
3. 10, 0, 8
4. 100, 5, 8
5. 10, 5, 8
6. 10, 8, 3

15. In a perceptron, let inputs be 0 or 1. Let the output be $y = 0$ or 1, and the correct output be C . The Perceptron Learning Rule can be summarised as:

1. If $y=0$ and $C=1$, then reduce threshold and reduce weights.
2. If $y=0$ and $C=1$, then increase threshold and reduce weights.
3. If $y=1$ and $C=0$, then reduce threshold and increase weights.
4. If $y=1$ and $C=0$, then increase threshold and increase weights.
5. If $y=0$ and $C=1$, then reduce threshold and increase weights.

16. We are learning to represent a function $Q(x)$. The input x is a 10 dimensional vector. Each dimension is an integer from 1 to 100. One approach to representing this function would be to have a "lookup table" or array of entries, where we have one entry $Q(x)$ for each input x . This array would be initially blank and we would fill it in by learning. The size of the array is:

1. 110
2. 1000
3. 10^{100}
4. 1010
5. 100^{10}
6. 100
7. 10

17. With large state-spaces, it is hard to draw the entire state-space representation of a problem. This is solved by:

1. It is not solved. We do not draw the state-space representation of the problem.
2. Writing a program to automatically draw it.
3. We draw a partial state-space representation of the problem.
4. Using symmetry to reduce the size of the state-space.

5. Doing a truncated search.
 6. Using a heuristic to reduce the size of the state-space.
18. Imagine the following in Best-first search. The heuristic evaluation function tells us a state is 4 steps away from the goal. We examine the children of this state and the heuristic evaluation function tells us all of them are 5 steps away from the goal. How can this happen?
1. This cannot happen.
 2. We are going the wrong way.
 3. We are in an infinite loop.
 4. The goal has been wrongly defined.
 5. The heuristic evaluation function is inaccurate.
 6. This always happens.
19. Neural Networks and GAs can both be seen as hill-climbing on an unknown landscape by taking a finite number of samples. Which of the following is the case:
1. Neural Networks make a directed move, GAs make a random move.
 2. Neural Networks make a random move, GAs make a random move.
 3. Neural Networks make a directed move, GAs make a directed move.
 4. Neural Networks make a random move, GAs make a directed move.
20. A problem has a *branching factor* of x . To examine all possible states between the current state and n steps into the future requires examining how many states?
1. n
 2. $x + x^2 + x^3 + \dots + x^n$
 3. x^x
 4. n^x
 5. $x + x^n$
 6. x
 7. x^n
 8. nx

Long questions

Answer 2 long questions out of 4. (25 marks each.) Total 50 marks.

- Each question gives **maximum number of sentences** for the answer. Do not exceed this.
- You might like to write your answers as **rough work** first to figure out what sentences to use. Make sure any rough work is crossed out.
- Put your sentences as **bullet points** not a paragraph.
- If you answer more than 2 questions, I will pick the best 2.

Question 1

[Total marks: 25]

- 1(a) We believe that some function exists from n -dimensional input x to output $f(x)$. That is, we can set up our system with input x and some process will give us a value for $f(x)$. We have no equation for f . [5 marks]
How can we *maximise* f , that is, find the value of x for which $f(x)$ is highest?
Maximum length: **3 sentences**
- 1(b) Consider a heuristic search as follows: [5 marks]
Initialise with 20 random solutions.
Repeat for many iterations:
{
 Test the 20 solutions.
 Pick the best n .
 Make small changes in them
 to generate 20 new solutions.
}
If $n=19$ what will this look like?
Maximum length: **2 sentences**
- 1(c) Consider trying to maximise a function of x , where x is an integer from 1 to 10^{20} . There are seven (and only seven) values of x where $f(x) = 10$. For all other x , $f(x) = 1$. The machine is not told this but has to find out by trial and error. [5 marks]
What would maximising this function by trial and error look like?
Maximum length: **3 sentences**
- 1(d) In normal software engineering, the engineer seeks to find the correct solution and code it as an algorithm. [5 marks]
In AI, the engineer may deliver a program that fails badly on occasion, and is pretty much known *not* to be the best solution.
How can this be justified?
Maximum length: **2 sentences**
- 1(e) Discuss trying to *maximise* the following function from exemplars. Explain what would happen during learning. [5 marks]
Input x is a number. It does *not* include the clock time.
The function is defined as follows:

```

if ( x > 10 )
    f(x) = 1  if clock time is PM.
    f(x) = -1 if clock time is AM.
else if ( x < -10 )
    f(x) = -1  if clock time is PM.
    f(x) = 1  if clock time is AM.
else f(x) = 0

```

Maximum length: **3 sentences**

Question 2

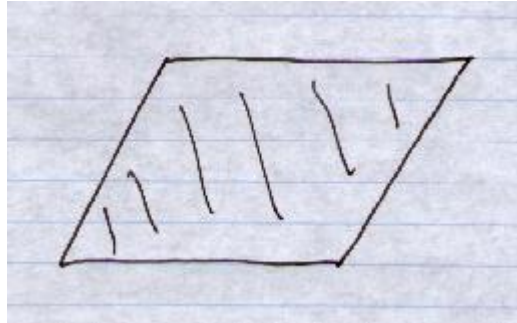
[Total marks: 25]

- 2(a) What are the problems in designing a heuristic evaluation function? [5 marks]
Maximum length: **2 sentences**
- 2(b) In X's and O's, we considered a heuristic of "Number of winning opportunities". You take the next move with the highest number of winning opportunities. You are done! No search needed. What is the catch? [5 marks]
Maximum length: **1 sentence**
- 2(c) In the N-puzzle problem, we use a heuristic (distance to goal estimate) as follows: $h(n)$ = number of pieces out of place. If $h(x) = 1$ then are we close to the goal? [5 marks]
Maximum length: **4 sentences**
- 2(d) When a problem is represented as a State Space Search problem, the "search tree" is drawn. True or false? [5 marks]
Maximum length: **1 sentence**
- 2(e) In the N-puzzle problem, we use a heuristic (distance to goal estimate) as follows: $h(n)$ = number of pieces out of place. If $h(x) = 5$ and $h(y) = 3$ then is y a better state than x ? [5 marks]
Maximum length: **2 sentences**

Question 3

[Total marks: 25]

- 3(a) Explain in English how you can make a multi-layer network that fires if and only if the input point is inside the parallelogram in this figure: [5 marks]



Maximum length: **2 sentences**

- 3(b) Consider the triangular area formed by the 3 lines: [5 marks]

$x=0$, $y=0$, $x+y=1$

Define the weights and thresholds of a *multi-layer* network that will fire only for points within the triangular area.

Maximum length: **6 sentences**

- 3(c) The following is the perceptron learning rule. We have multi-dimensional input, where each component of the input l_i is binary 0 or 1. It produces output $y = 0$ or 1. We show it the "correct" output $\mathbf{O} = 0$ or 1. We modify the weights and the threshold by the rules: [5 marks]

$$w_i := w_i + C (\mathbf{O} - y) l_i$$

$$t := t - C (\mathbf{O} - y)$$

where C is some positive constant.

Question: What happens if y is too large?

Maximum length: **1 sentence**

- 3(d) A mobile robot has Input state = (angle of direction of nearest wall, velocity value). Output behaviour = one of (turn left, turn right, stop, keep going). [5 marks]

The Input state is clearly missing some information. What are the consequences of this missing information?

Maximum length: **3 sentences**

- 3(e) We are trying to build a character recognition program. [5 marks]
Characters are images of 100×100 pixels. Each pixel is black or white. We are trying to recognise the digits "0" to "9".
Input is 2 dimensional so therefore the input vector should be 2 dimensional. True or false?

Maximum length: **2 sentences**

Question 4

[Total marks: 25]

- 4(a) In a genetic algorithm (GA), let the probability of an individual being picked for reproduction be: [5 marks]

$$p = \frac{\text{my fitness}}{\text{sum of fitnesses of all creatures}}$$

sum of fitnesses of all creatures

Fitness cannot be both positive and negative. Why?

Maximum length: **1 sentence**

- 4(b) In a genetic algorithm: What would happen if rather than picking the fittest, you define a *minimum* fitness that individuals have to reach in order to reproduce? [5 marks]

Maximum length: **1 sentence**

- 4(c) In a genetic algorithm (GA), let p_m be the probability that any bit in the chromosome mutates at the point where it is inherited from the parent by the child. Let the chromosome be binary, so that mutation means a bit flip. [5 marks]

What would happen if $p_m = 1$?

Maximum length: **4 sentences**

- 4(d) In a genetic algorithm: What would happen if only 1 in a billion solutions has non-zero fitness? [5 marks]

Maximum length: **2 sentences**

- 4(e) In a genetic algorithm (GA), let each individual i have fitness $f(i)$, and let its probability of reproducing, $p(i)$, be: [5 marks]

$$p(i) = \frac{e^{\frac{f(i)}{T}}}{\sum_j e^{\frac{f(j)}{T}}}$$

where T is the "temperature".

Consider the individuals:

i	1	2	3
$f(i)$	-2	-2	-1

What happens as $T \rightarrow 0$?

Maximum length: **4 sentences**

Bonus question

Bonus question. (10 marks.)

It is "bonus" because I tell you it in advance. The question is to make you do some reading. You can prepare this question in advance.

Bonus question:

Tell me about some AI-related algorithm or software that was *not* mentioned on the course.

CA318 - Advanced Algorithms and AI Search
Answer sheet

Black out the square that matches the correct answer.

Exam No: _____

Class: _____

Seat No: _____

Question	1.	2.	3.	4.	5.	6.	7.	8.
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